

Noise Analysis — 4um

PAGE 1 — FILE STATISTICS

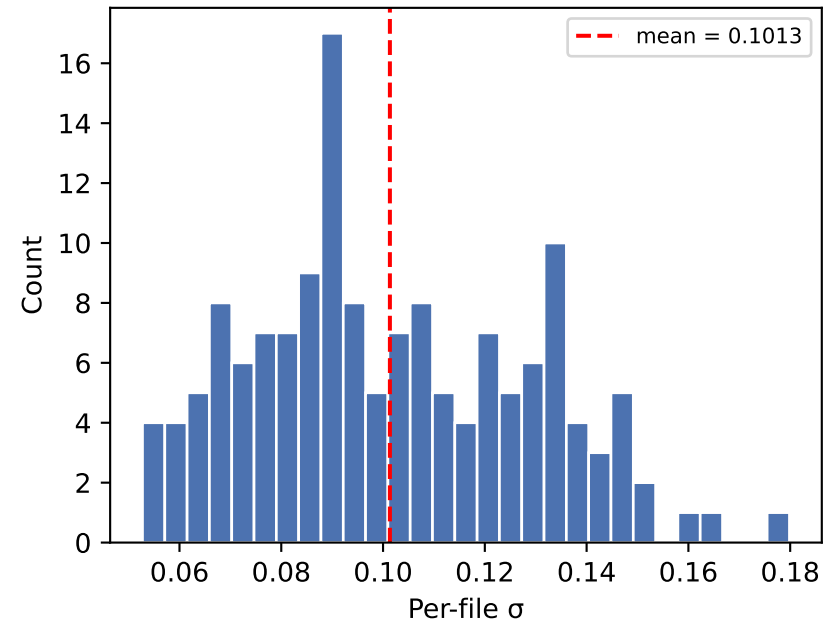
Folder: data/dataset_hamming_c1/test/4um
Files: 149
Signal length: 16384–16384 samples
Dtype: float64

Aggregate noise statistics:

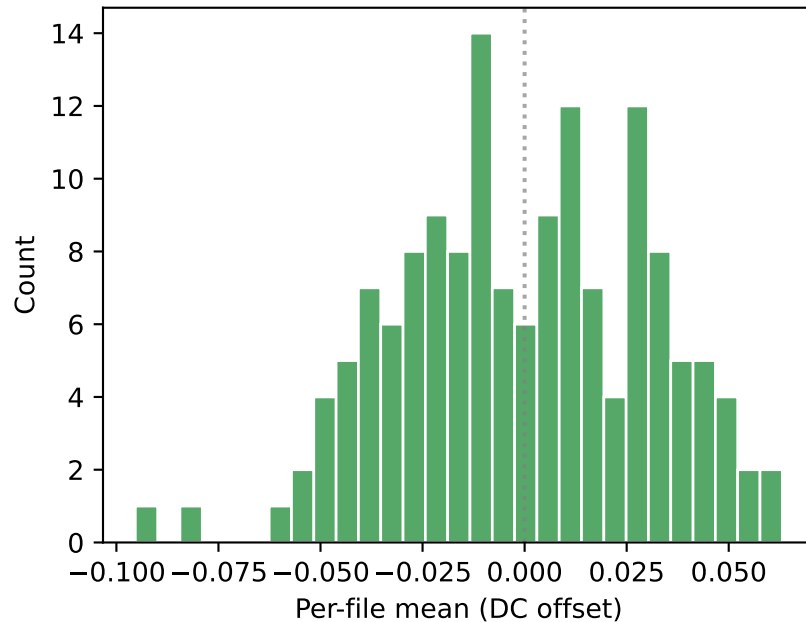
mean(σ): 0.101345
std(σ): 0.027332
CV(σ): 0.2697
mean(RMS): 0.105985
mean(kurtosis): 3.3785 (σ = Gaussian)

mean(DC): -0.000694
std(DC): 0.030575
Amplitude range: [-1.0532, 1.0354]

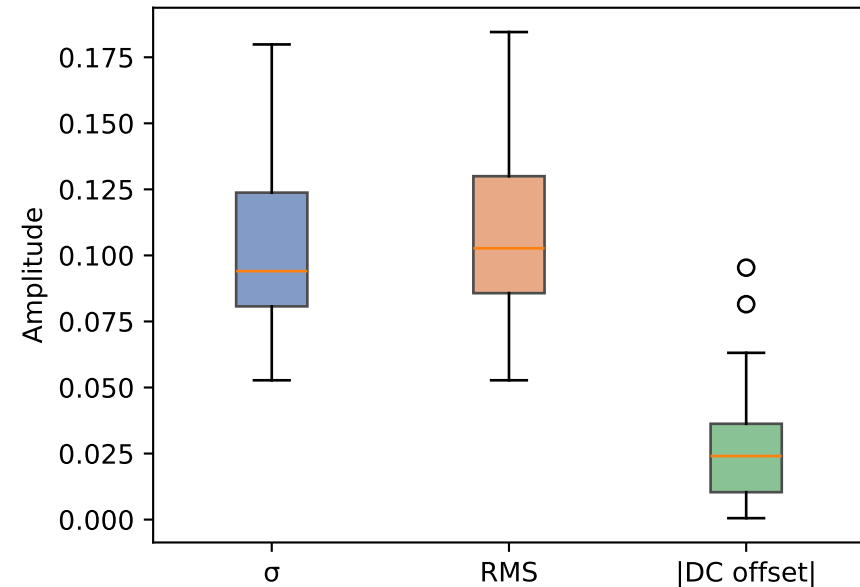
Distribution of per-file standard deviations



Distribution of per-file DC offsets

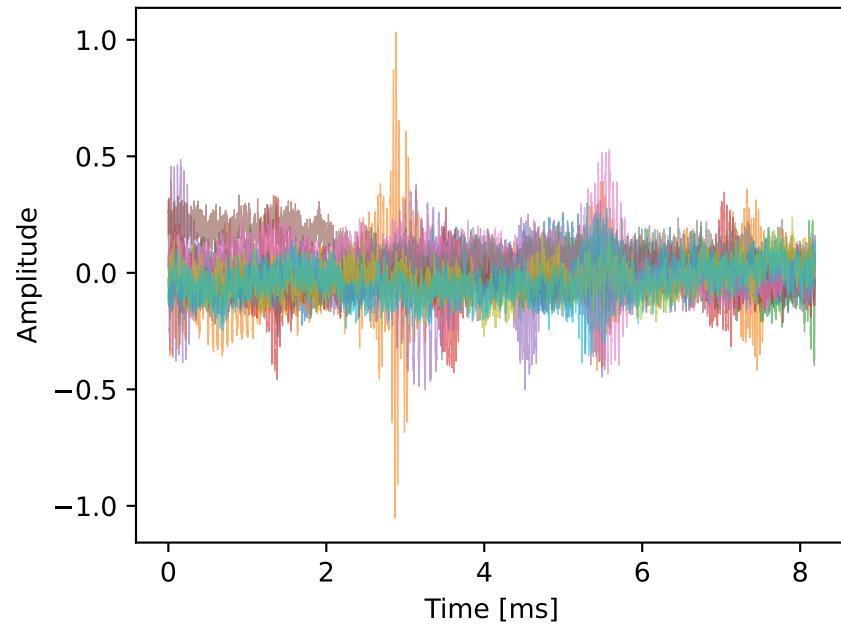


Distribution of amplitude metrics

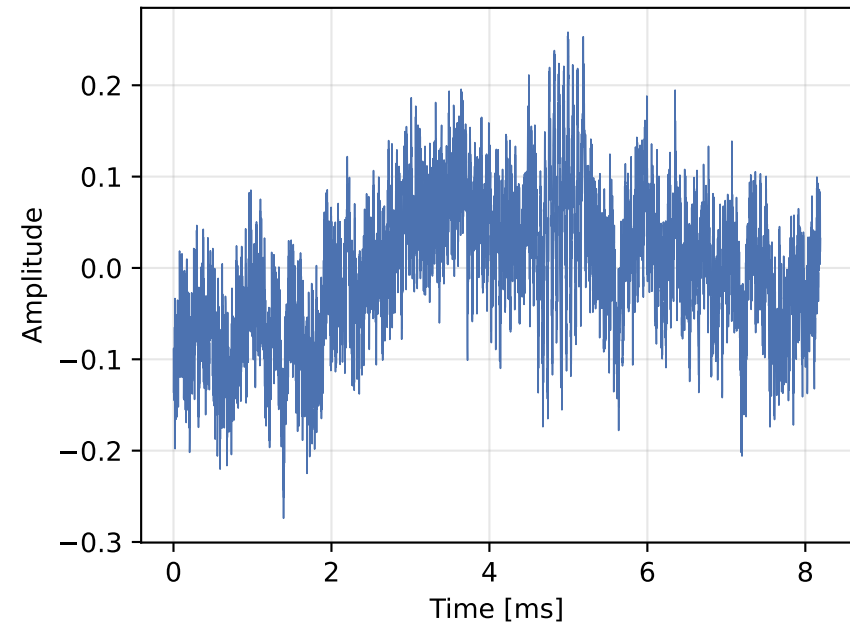


Time-Domain Analysis — 4um

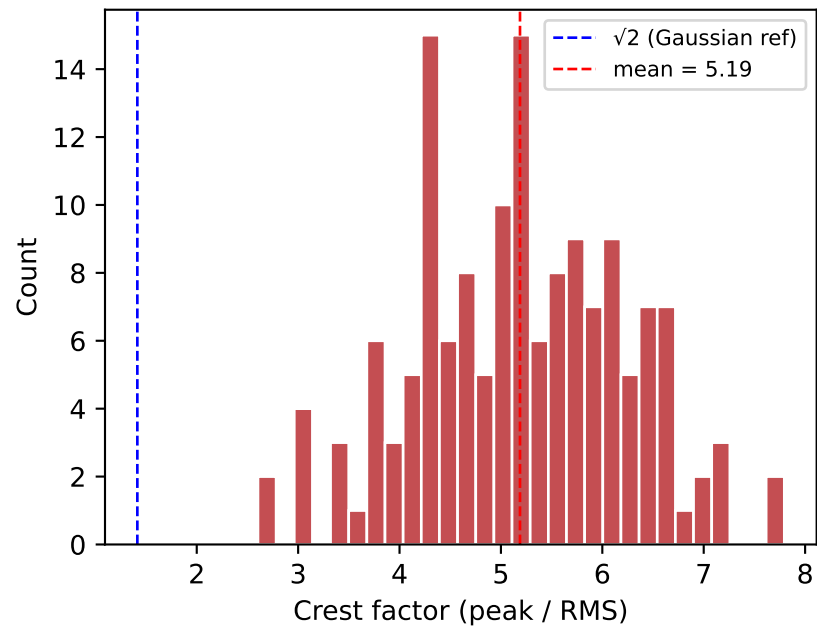
Overlay of 10 random signals



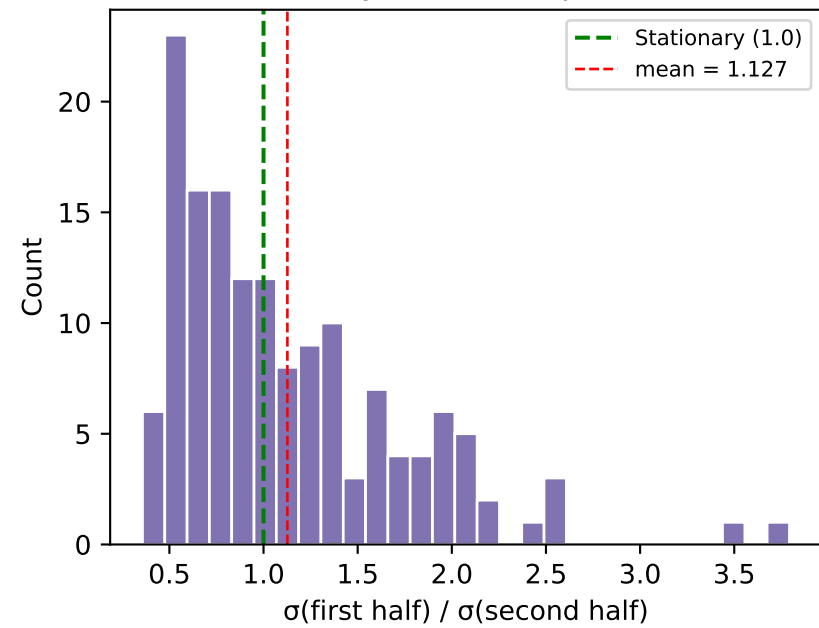
Signal #145 (detail)



Crest factor distribution

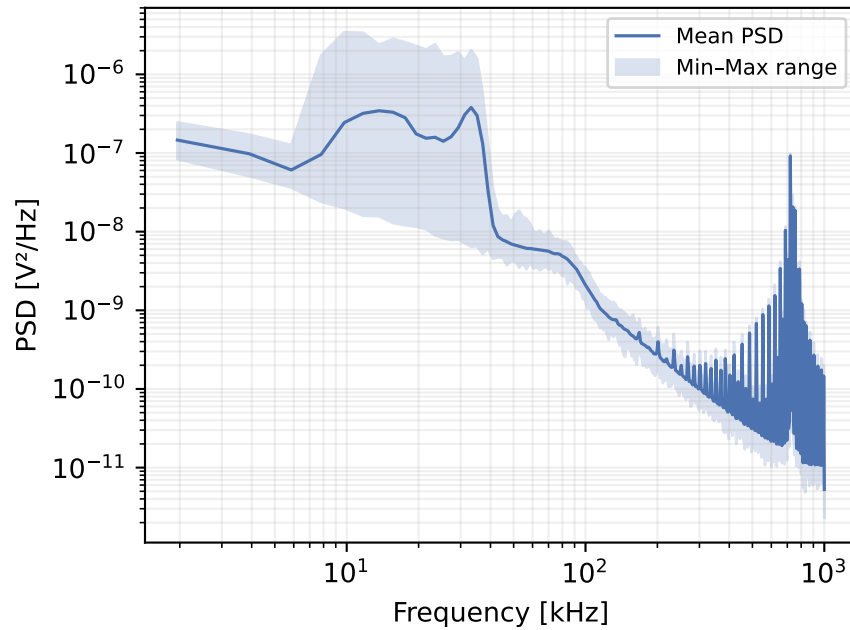


Stationarity test (half-split σ ratio)

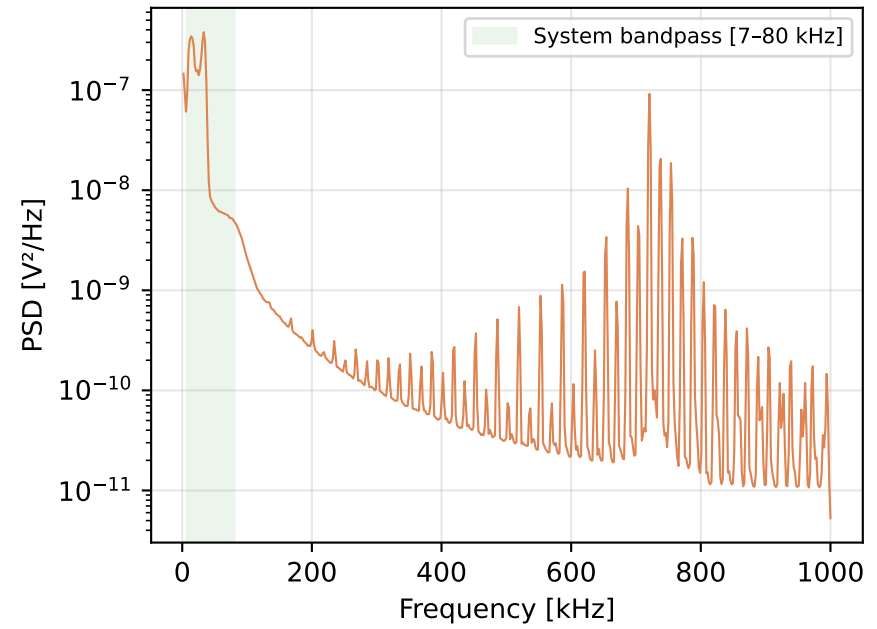


Power Spectral Density — 4um

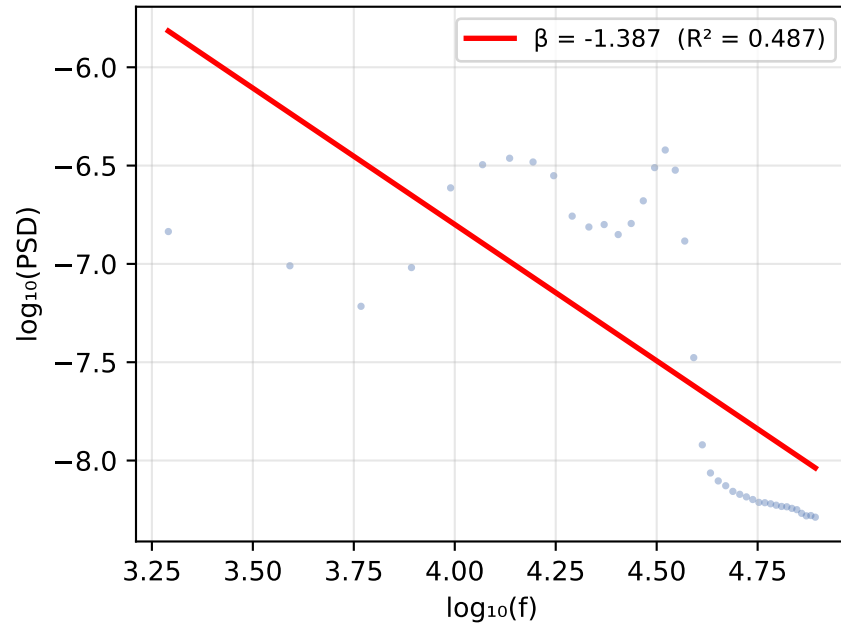
PSD (log-log)



PSD (linear freq, log power)



Spectral slope fit (1-80 kHz)



SPECTRAL ANALYSIS SUMMARY

Spectral slope β : -1.3868
Fit quality R^2 : 0.4872
Spectral flatness: 0.0180 (1.0 = white)

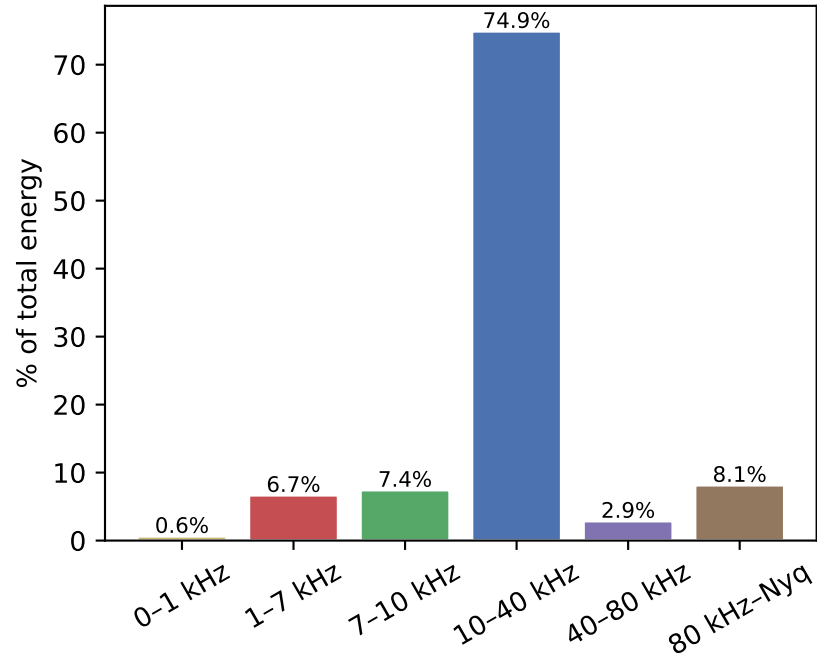
Slope interpretation:

$\beta \approx 0 \rightarrow$ White noise (flat PSD)
 $\beta \approx -1 \rightarrow$ Pink / $1/f$ (equal energy per octave)
 $\beta \approx -2 \rightarrow$ Brown / $1/f^2$ (random walk)

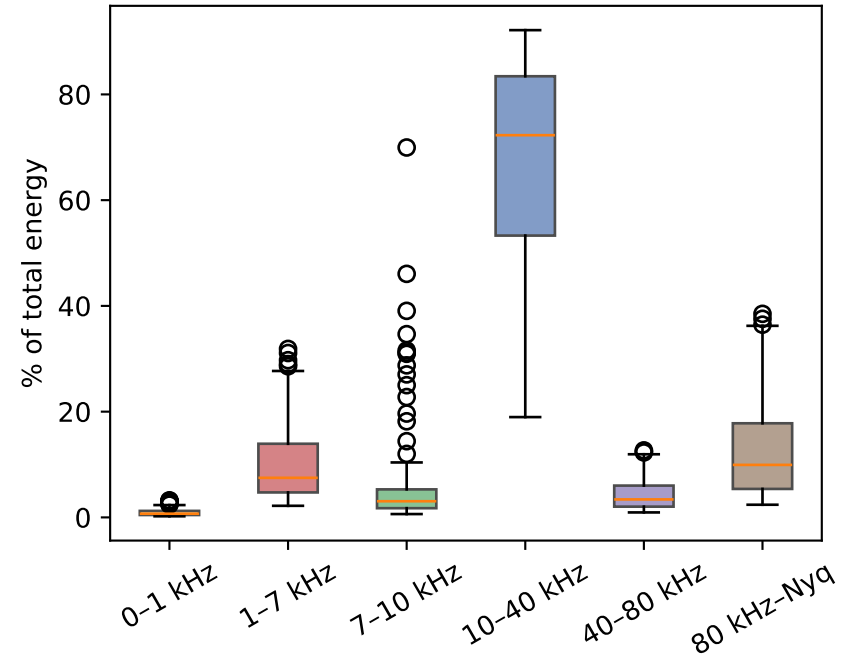
Classification: Pink ($1/f$)

Frequency Band Energy — 4um

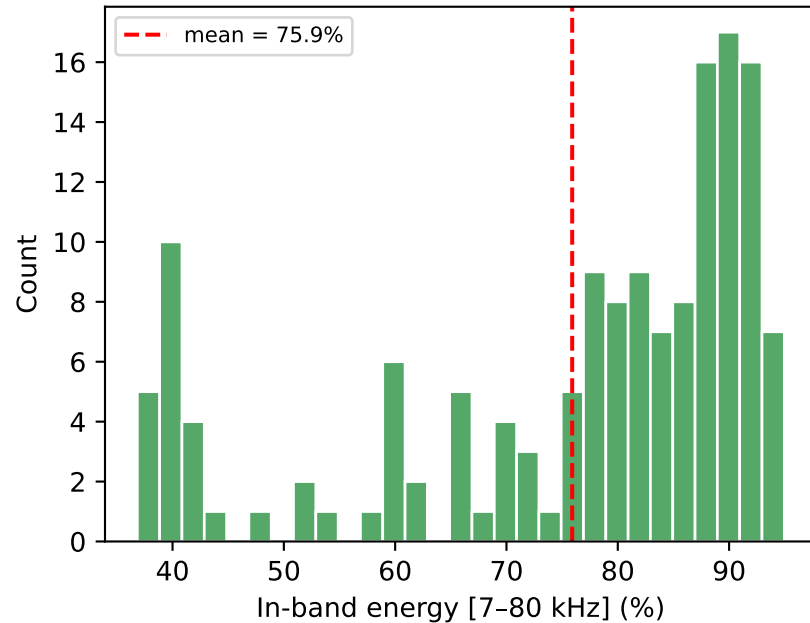
Average energy per frequency band



Per-file band energy distribution



System bandpass energy ratio per file



BAND ENERGY SUMMARY

Band	Mean %	Std %
0-1 kHz	0.6%	0.7%
1-7 kHz	6.7%	7.6%
7-10 kHz	7.4%	9.4%
10-40 kHz	74.9%	22.3%
40-80 kHz	2.9%	3.3%
80 kHz-Nyq	8.1%	10.1%
In-band [7-80k]	75.9%	17.5%

Noise Type Classification — 4um

NOISE TYPE CLASSIFICATION

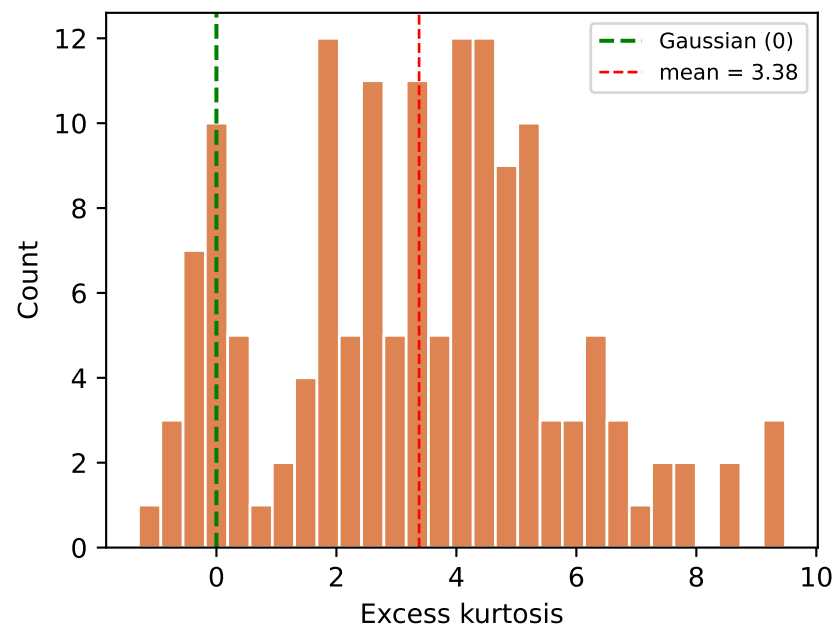
Spectral color: Pink (1/f)
slope β = -1.3868
flatness = 0.0180

Amplitude dist.: Non-Gaussian
kurtosis (mean): 3.3785
(excess kurtosis: 0 = Gaussian,
>0 = heavy-tailed, <0 = light-tailed)

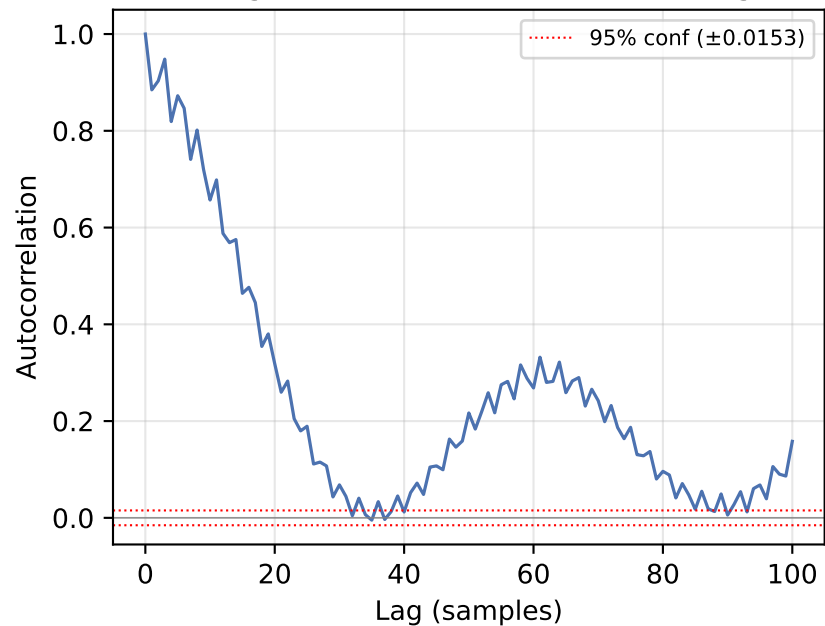
Decision boundaries:
 $|\beta| < 0.3$ → White
0.3–1.5 → Pink (1/f)
> 1.5 → Brown (1/f²)

$|\text{kurt}| < 0.5$ → Gaussian
0.5–2.0 → Near-Gaussian
> 2.0 → Non-Gaussian

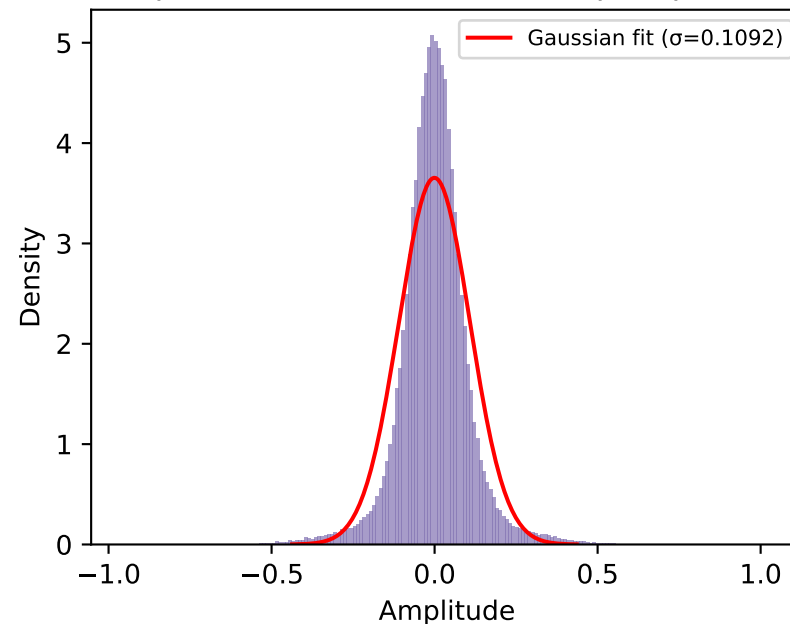
Per-file kurtosis distribution



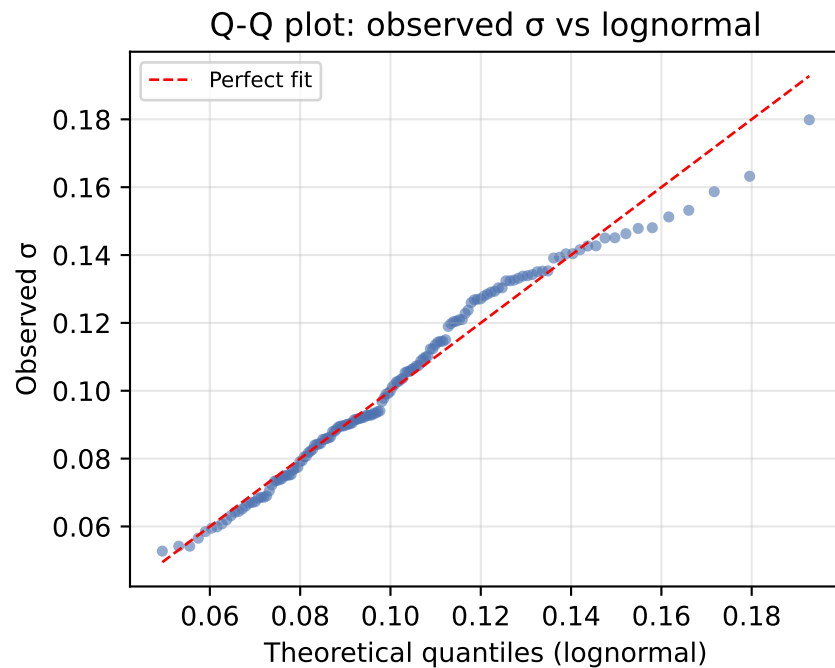
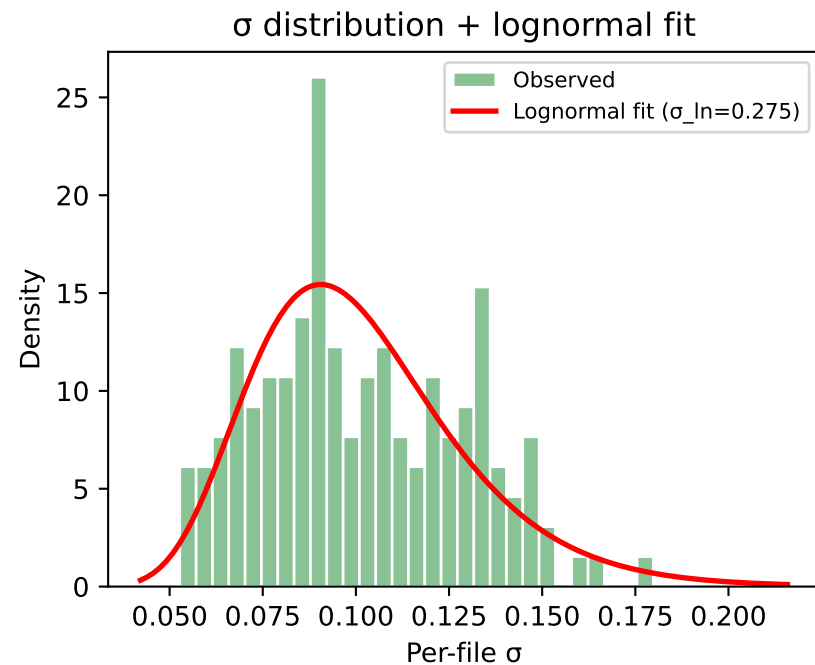
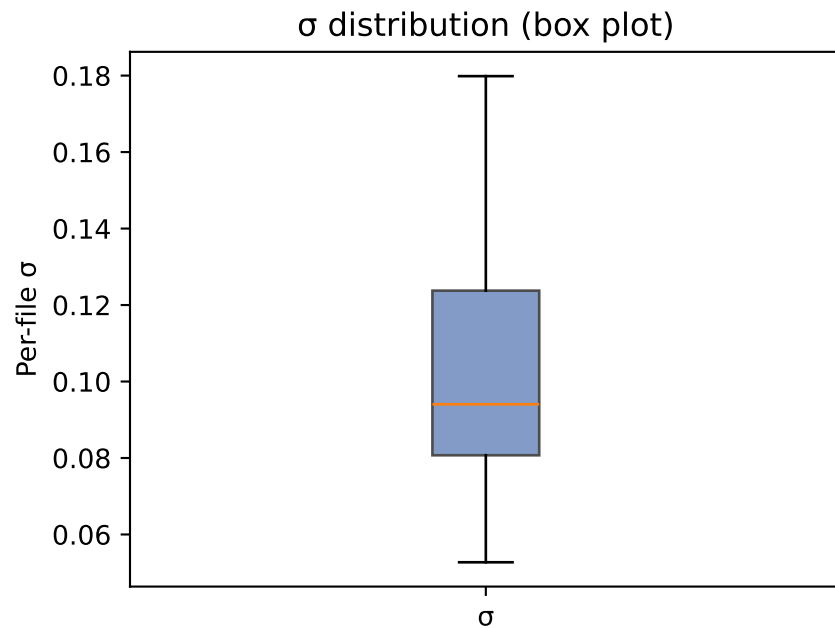
Average autocorrelation (first 100 lags)



Amplitude distribution (all samples pooled)



Amplitude Variability (Inter-file) — 4um



AMPLITUDE VARIABILITY SUMMARY

Number of files: 149
mean(σ): 0.101345
std(σ): 0.027332
CV = std(σ)/mean(σ): 0.2697

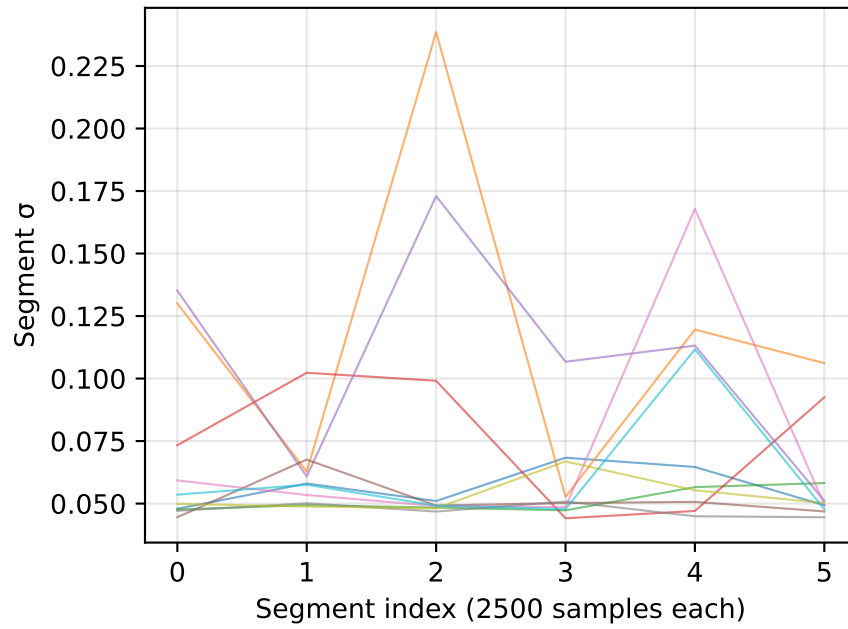
Lognormal fit parameters:
shape ($\sigma_{\ln}$): 0.2748
scale ($\exp(\mu_{\ln})$): 0.097657

Reference (from real noise files):
CV_ref = 0.19
Ratio CV/CV_ref: 1.42

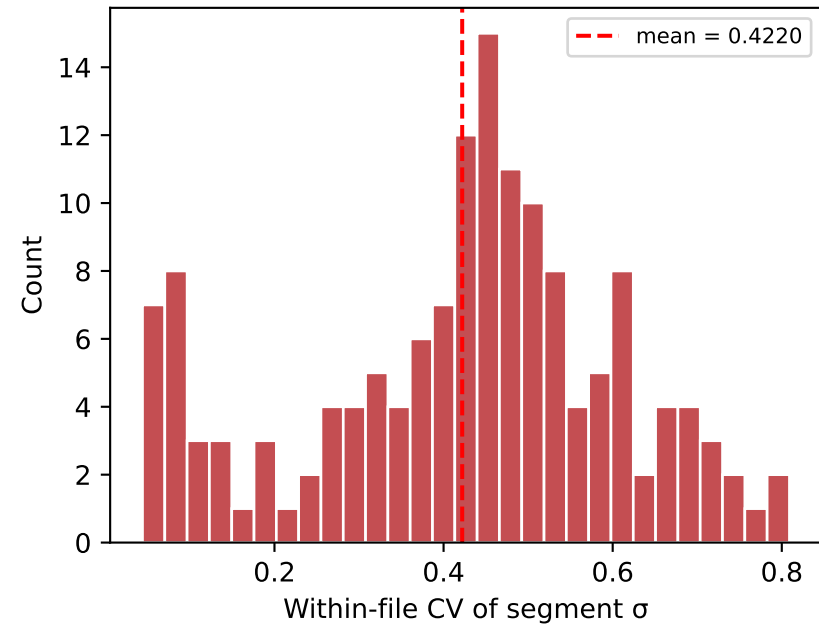
Interpretation:
High variability – mixed conditions or signal content

Segment-Level Analysis — 4um

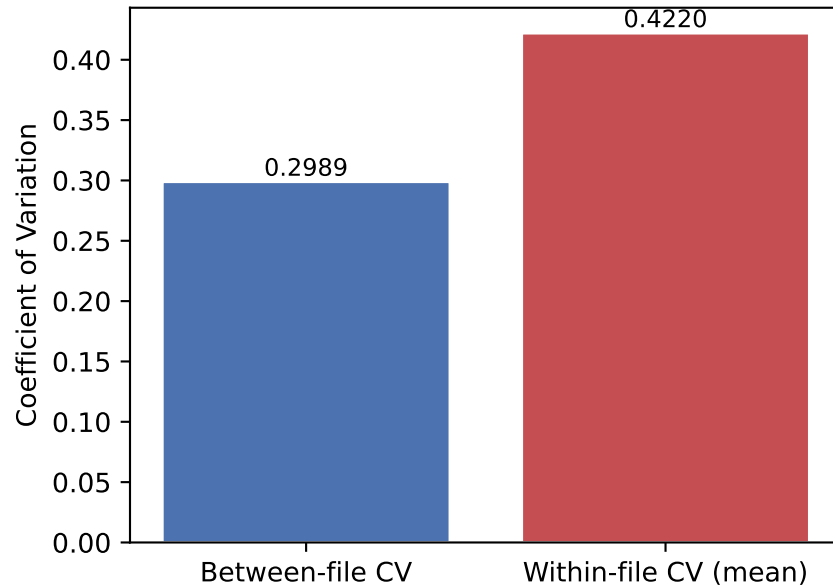
σ evolution within files (10 shown)



Within-file noise stationarity



Between-file vs within-file variability



SEGMENT ANALYSIS SUMMARY

Segment length: 2500 samples
Files analyzed: 149 / 149
Segments per file: 6–6

Between-file CV: 0.2989
Within-file CV (mean): 0.4220
Within-file CV (std): 0.1866

Interpretation:

Significant non-stationarity – noise level varies within individual recordings
Within-file variability is comparable to between-file variability